Chuanyang Jin

Email | Personal Website | Github | Linkedin

Research Interests

Enhancing the robustness, truthfulness, and instruction-following capability of foundation models, such as multimodal models and LLMs. Leveraging foundation models to develop agents that excel in long-term decision making by effectively planning, reasoning, leveraging memory, and learning from real-world feedback.

Education

New York University Sep 2020 – May 2024

B.A in Computer Science with Honors; B.A in Mathematics *GPA: 3.96*

- Relevant coursework: Deep Learning (Grad), Machine Learning (Grad), Natural Language Processing (Grad), Computer Vision (Grad), Algorithms (TA), Robot Intelligence, Numerical Analysis, Differential Equations

Research Experience

Massachusetts Institute of Technology May 2023 – Now

Visiting Researcher, advised by Joshua B. Tenenbaum, Antonio Torralba, Tianmin Shu *Boston*

New York University Jan 2022 – Now

Research Assistant, advised by Oded Regev, Saining Xie *New York*

Publications / Preprints

- **C. Jin**, Y. Wu, J. Cao, J. Xiang, Y.-L. Kuo, Z. Hu, T. Ullman, A. Torralba, J. B. Tenenbaum, T. Shu. MMToM-QA: Multimodal Theory of Mind Question Answering. *In Submission to ICLR 2024.*
- K. Jha, T. A. Le, **C. Jin**, Y.-L. Kuo, J. B. Tenenbaum, T. Shu. Neural Amortized Inference for Nested Multi-agent Reasoning. *Oral Presentation at AAAI Summer Symposium. In Submission to AAAI 2024. arXiv:2308.11071.*
- J. Wu*, Y. Xiao*, Y. Liu*, L. Wen*, **C. Jin**, S. Paul, C. He, O. Regev, J. Fei. Deciphering RNA Localization in Nuclear Speckles. *Manuscript available upon request.*
- **C. Jin**. Self-Supervised Image Captioning with CLIP. *In Submission to NeurIPS 2023 Workshop on Self-Supervised Learning. arXiv:2306.15111.*
- P. Liu, **C. Jin**, Z. Zhong. Efficient Video Prediction and Segmentation with Sparse Labeled Data. *In Submission to NeurIPS 2023 Workshop on Distribution Shifts.*
- **C. Jin***, S. Zhang*, T. Shu, Z Cui*. The Cultural Psychology of Large Language Models: Is ChatGPT a Holistic or Analytic Thinker? *In Submission to PNAS. arXiv:2308.14242.*

Research Projects

MMToM-QA: Multimodal Theory of Mind Question Answering June 2023 – Oct 2023

Advisor: Joshua B. Tenenbaum, Antonio Torralba, Tianmin Shu

- Introduce a multimodal Theory of Mind question answering (MMToM-QA) benchmark through procedural generation, which comprehensively evaluates the cognitive ability to understand people's minds (e.g., beliefs, goals) both on multimodal data and on different kinds of unimodal data in rich and diverse scenarios.
- Propose BIP-ALM (Bayesian Inverse Planning Accelerated by Language Models) to engineer multimodal ToM, which extracts, aligns, and fuses symbolic representations from multimodal data, formulates an agent's behavior as a forward generative model, and finetunes language models to amortize the policy for scalable inverse planning.
- Conduct a systematic comparison of human performance, BIP-ALM, state-of-the-art LLMs, and multimodal models. Demonstrate that current ML models lack robust ToM capacity, while BIP-ALM shows promising results.
- Submit a paper (as the 1st author) to ICLR 2024.

Neural Amortized Inference for Nested Multi-agent Reasoning May 2023 – Aug 2023

Advisor: Joshua B. Tenenbaum, Tianmin Shu

- Introduce a novel approach that recursively performs particle sampling using recognition networks and leveraging data from generative models to amortize high-order multi-agent reasoning and emulate human social inference.

- Construct two challenging multi-agent test scenarios in autonomous driving and construction domains, and demonstrate that our method is computationally efficient while exhibiting minimal degradation in accuracy.
- Submit a paper to AAAI 2024.

Optimization of Diffusion Transformers Feb 2023 – May 2023

Advisor: Prof. Saining Xie

- Implement optimizations such as pre-extracted VAE features, improved attention mechanism, gradient checkpointing, and mixed precision training, achieving a 95% speed increase and 60% memory reduction.
- Investigate 4-bit post-training quantization of diffusion modes and progressive model distillation for fast sampling.
- Open-source the project, receiving over 200 GitHub Stars and subsequent integration into popular repositories.

Deciphering RNA Localization in Nuclear Speckles May 2022 – Oct 2023

Advisor: Prof. Oded Regev

- Analyze speckle-enriched transcripts and reveal sequence logics that drive RNA localization to nuclear speckles.
- Develop an extensive suite of tools for the analysis of experimental data, utilizing interpretable machine learning to decipher complex structures and interactions spanning over 30 factors, coupled with compelling visualizations.
- Uncover two sets of sequence features and propose a model where nuclear speckles kinetically coordinate splicing.
- Co-author a paper and showcase findings at the NYU Undergraduate Research Conference.

Visual Place Recognition Using Perspective Images Jan 2022 – May 2022

NYU AI4CE Lab

- Enhance a visual place recognition model to accommodate perspective images along with panoramas, by leveraging an encoder-decoder model to perform scene completion. Finetune it via a pretext task of relative pose estimation.
- Conduct experiments on simulated and real datasets, outperforming baselines in recall rate and robustness.
- Develop a unified framework that handles various supervision methods, and both point cloud and RGB-D images.

Self-Supervised Image Captioning with CLIP May 2023 – July 2023

- Propose a self-supervised captioning method by maximizing the CLIP relevance between images and captions.
- Leverage CLIP encoding, mapping network, and LLaMA 2, bolstered by advanced initialization and optimization.
- Outperform SOTA method while only using 2% labeled data in both reference-based and reference-free metrics.
- Submit a paper (as the 1st author) to NeurIPS 2023 Workshop on Self-Supervised Learning.

Efficient Video Prediction and Segmentation with Sparse Labeled Data April 2023 – May 2023

- Propose a pseudo labeling and transfer learning method with an enhanced SimVP model for end-to-end video prediction and segmentation, effectively addressing the challenge of scarce labeled data and distribution shifts.
- Achieve substantial improvement over baseline methods in both Jaccard Index and MAE across multiple tasks.
- Submit a paper to NeurIPS 2023 Workshop on Distribution Shifts.

Work / Projects

Software Development Intern June 2021 – Aug 2021

Chinasoft International Ltd. *Nanjing*

- Maintain and extend the functionalities of a company's personnel/financial data management web platform.
- Design, implement, and test tools using MySQL to enable administrators to operate on shared databases.
- Leverage Django web framework, JavaScript, and HTML/CSS to create a user-friendly interactive interface.

AutoCorp: A Fully Autonomous Company April 2023 - May 2023

- Launch AutoCorp with LLMs as a core controller to generate reasoning traces and perform sequential actions.
- AutoCorp generates creative ads, branding, and product designs using GPT-4/DALL-E/ControlNet, interacts with customers, manages memory, learns from feedback, and performs long-term reasoning and automatic updates.

AutoLink: A Tool for Strategic Networking March 2023 - April 2023

- Develop a LinkedIn plugin integrating LLMs, data enrichment tools, graph-based search, and behavior analytics.
- AutoLink promotes strategic networking by profile information extraction, relevance-based ranking of potential connections/opportunities, personalized message generation, and adaptive learning from user preferences.

Class Embedding Patterns in Classifier-Free Diffusion Oct 2022 – Dec 2022

- Explore approaches to generate mappings from class labels and integrate them into the reverse diffusion process.
- Design and implement class label embedding patterns such as Uniform, Pyramid, and Bottleneck Embedding.

- Evaluate their performance, properties, and trade-offs. Leverage curriculum learning to handle noisy labels.

Parallel Command Processor Sep 2022 – Dec 2022

- Build a command interpreter that handles I/O redirection and pipe, and manages background/foreground process.
- Implement a thread pool using POSIX threading to parallelize provided tasks for improved performance.

Graph Visualization of Matrices March 2022 – May 2022

- Introduce a novel shift method to speed up the convergence of power iterations for computing tensor eigenpairs.
- Implement dimension reduction with power iterations and propose an innovative *graph Laplacian* concept for matrix visualization. Leverage them to create a toolbox that helps uncover structures in complex real-world relations.

Gene Mutation Detection Platform May 2019 – May 2021

- Propose a mutation detection algorithm for Sanger sequencing data, utilize computational geometry for feature extraction, employ deep learning for mutation localization and classification, and develop visualization tools.
- Launch an online platform adopted by leading hospitals to reduce diagnosis periods and receive a $10k reward.

Honors/Awards

- NYU Computer Science Prize for the Most Promising Junior Student (Top 1/300) 2023
- Finalist & MAA Award & COMAP Scholarship (Top 4/27205), Interdisciplinary Contest in Modeling 2022
- Champion (Top 1/51), Tsinghua X-lab AI Hackathon 2021
- Distinguished Honor Roll (Top 1%), AMC12 & USAMO qualifier 2020
- International Bronze Award (Top 3/500), Shing-Tung Yau High School Computer Science Award 2019
- Finalist (Top 8/3790), FIRST Robotics Competition World Championship 2019
- First Prize (Top 0.1%), China National Mathematical Olympiad for Senior High School 2018

Services/Activities

- Influential writer on Zhihu Q&A forum with 26K followers and 16,000K total reads 2020 - Present
- Teaching Assistant of CSCI-UA 310 Basic Algorithms Fall 2022
- Executive board member of NYU Mathematics Society 2021 - 2022
- Director of Environmental Sustainability Committee at NYU Shanghai 2020 - 2021

Skills

- Language: Python, C++, C, Java, JavaScript, Matlab, SQL, R, Octave, Linux, Git
- Framework: PyTorch, TensorFlow, JAX, HuggingFace Transformers, Scikit-learn, OpenCV